2A Operations with Positive and Negative Fractions

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Evaluate each fraction addition or subtraction in simplest form:

(a) $-\frac{5}{8}+\frac{3}{4}$ (b) $-\frac{2}{3}-\frac{4}{5}$ (c) $\frac{5}{6}-\left(-\frac{7}{9}\right)$ (d) $-2\frac{1}{4}+1\frac{2}{3}$

Q2. Evaluate each fraction multiplication in simplest form:

(a) $-\frac{4}{9}\times\frac{3}{8}$ (b) $-\frac{5}{12}\times\left(-\frac{9}{10}\right)$ (c) $-1\frac{3}{5}\times\frac{15}{16}$ (d) $-2\frac{1}{3}\times\left(-1\frac{2}{7}\right)$

Q3. Evaluate each fraction division in simplest form:

(a) $-\frac{7}{10}\div\frac{14}{15}$ (b) $-\frac{8}{9}\div\left(-\frac{4}{15}\right)$ (c) $1\frac{4}{5}\div\left(-\frac{9}{10}\right)$ (d) $-3\frac{1}{2}\div\left(-2\frac{1}{3}\right)$

Q4. Evaluate each multi-step fraction expression:

(a) $-\frac{1}{2}+\frac{3}{4}\times\left(-\frac{8}{9}\right)$ (b) $\left(-\frac{2}{3}\right)^{2}-\frac{7}{9}$ (c) $\left(\frac{1}{4}-\frac{5}{6}\right)\div\left(-\frac{7}{12}\right)$ (d) $\left(-\frac{3}{2}\right)^{3}+\frac{5}{8}$

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. A bushwalking track in the Grampians is $14\frac{2}{5}\ \text{km}$ long. After walking $\frac{5}{8}$ of the total distance, how many kilometres of the track remain?

Q6. A rainwater storage tank in Gippsland is $\frac{4}{5}$ full. After $\frac{1}{3}$ of the total tank capacity is used for garden irrigation and $\frac{3}{10}$ of the total capacity is added by weekend rain, what fraction of the tank is now full?

Q7. A carpenter has a length of Tasmanian oak timber measuring $6\frac{3}{4}\ \text{m}$. How many shelf brackets each requiring $\frac{3}{8}\ \text{m}$ of timber can be cut from this length?

Q8. During a science experiment, the temperature of a chemical solution changes by $-1\frac{3}{4}^{\circ}\text{C}$ every minute for 6 minutes. If its initial temperature was $8\frac{1}{2}^{\circ}\text{C}$, what is its final temperature?

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Find the rational number that lies exactly halfway (the mean) between $-\frac{5}{6}$ and $-\frac{1}{4}$. Show your working clearly.

Q10. Explain why dividing by a negative proper fraction between -1 and 0 (such as $-\frac{2}{5}$) always produces a result with a larger magnitude than the original number, and give a numerical example.